

Homework Chapter One

Hw3

$$X_n = \begin{cases} 1 & n=0,1,2,3,4,\dots \\ 3 \end{cases}$$

State space = $\{1, 3\}$

Index set $T = \{0, 1, 2, 3, 4, \dots\}$

Markov Property:

$$P(X_{n+1}=j | X_n=i) = P(X_{n+1}=j | X_n=i, X_{n-1}=i_{n-1}, \dots, X_0=i_0)$$

(b) Transition matrix

$$P(X_{n+1}=3 | X_n=1) = 2/3 = p_{13}$$

$$P(X_{n+1}=1 | X_n=3) = 3/5 = p_{31}$$

$$P(X_{n+1}=1 | X_n=1) = 1/3 = p_{11}$$

$$P(X_{n+1}=3 | X_n=3) = 2/5 = p_{33}$$

Initial distribution

$$\pi_1 = P(X_0=1)$$

$$\pi_3 = P(X_0=3)$$

$$P = \begin{pmatrix} p_{11} & p_{13} \\ p_{31} & p_{33} \end{pmatrix} = \begin{pmatrix} 1/3 & 2/3 \\ 3/5 & 2/5 \end{pmatrix}$$

Hw4

$$X_n = \begin{cases} 0 & n=0 \\ 1 & n=1 \\ 2 & n=2 \end{cases}$$

$$S = \{0, 1, 2\}$$

$$\pi_0 = P(X_0=0) = 0.2$$

$$\pi_1 = P(X_0=1) = 0.5$$

$$P(X_{n+1}=0 | X_n=0) = p_{00} = 0.5$$

$$P(X_{n+1}=1 | X_n=0) = p_{01} = 0.4$$

$$P(X_{n+1}=2 | X_n=0) = p_{02} = 0.1$$

$$P(X_{n+1}=0 | X_n=1) = p_{10} = 0.3$$

$$P(X_{n+1}=1 | X_n=1) = p_{11} = 0.4$$

$$P(X_{n+1}=2 | X_n=1) = p_{12} = 0.3$$

$$P(X_{n+1}=0 | X_n=2) = p_{20} = 0.2$$

$$P(X_{n+1}=1 | X_n=2) = p_{21} = 0.3$$

$$P(X_{n+1}=2 | X_n=2) = p_{22} = 0.5$$

$$\begin{aligned} P(X_2=2) &= P(X_1=2, X_0=0) + P(X_1=2, X_0=1) + P(X_1=2, X_0=2) \\ &\Rightarrow p_{02} P(X_0=0) + p_{12} P(X_0=1) + p_{22} P(X_0=2) \\ &\Rightarrow (0.1)(0.2) + (0.3)(0.5) + (0.5)(0.3) \\ &\Rightarrow 0.32 \end{aligned}$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$P(X_2=1) = P(X_1=1, X_0=0) + P(X_1=1, X_0=1) + P(X_1=1, X_0=2)$$

$$\begin{aligned}
 & 3) P(X_2=2, X_3=0, X_4=1) \\
 & \rightarrow P(X_4=1 | X_3=0) P(X_3=0 | X_2=2) P(X_2=2, X_1=1) \\
 & \rightarrow P_{01} \cdot P_{00} \cdot P_{02} P(X_1=0) + P_{01} P_{20} P_{12} P(X_1=1) + P_{01} P_{20} P_{22} P(X_1=2) \\
 & \rightarrow 0.00248 + 0.00888 + 0.0128 \\
 & \rightarrow 0.02416
 \end{aligned}$$

HW 5

$$\begin{aligned}
 & P_{ij} \in [0, 1], \sum_{j=1}^{\infty} P_{ij} = 1 \\
 & \sum_{i=1}^{\infty} P_{ij} = 1
 \end{aligned}$$

$$a) P_{ij} = \left(\frac{1}{2}\right)^j, \quad i, j = 1, 2, 3, \dots$$

$$j=1 \Rightarrow P_{i1} = \frac{1}{2}$$

$$j=2 \Rightarrow P_{i2} = \left(\frac{1}{2}\right)^2$$

$$j=3 \Rightarrow P_{i3} = \left(\frac{1}{2}\right)^3$$

$$j=4 \Rightarrow P_{i4} = \left(\frac{1}{2}\right)^4$$

$$0 \leq P_{ij} \leq 1$$

$$\sum_{j=1}^{\infty} P_{ij} = \sum_{j=1}^{\infty} \left(\frac{1}{2}\right)^j = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$$

$$q = \frac{1}{2}, r = \frac{1}{2} \quad \{\text{Geometric series}\}$$

$$\sum_{j=1}^{\infty} a r^{j-1} = \frac{a}{1-r} = \frac{\frac{1}{2}}{1-\frac{1}{2}} = 1$$

$$\sum_{j=1}^{\infty} P_{ij} = 1$$

$$b) P_{ij} = \left(\frac{1}{2}\right)^j, \quad i, j = 1, 2, 3, \dots$$

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \dots \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \dots \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

$$c) P(X_i=j) = P(X_i=j, X_0=1) + P(X_i=j, X_0=2) + \dots$$

$$\rightarrow \sum_{k=1}^{\infty} P(X_i=j | X_0=k) P(X_0=k)$$

$$\rightarrow \sum_{k=1}^{\infty} P(X_i=j | X_0=k) P(X_0=k)$$

$$\rightarrow \sum_{k=1}^{\infty} P_{ij} P(X_0=k)$$

$$\rightarrow \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^j \left(\frac{1}{2}\right)^k = \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^{j+k}$$

$$\rightarrow 1 \cdot \left(\frac{1}{2}\right)^j$$

$$P(X_i=j) = \left(\frac{1}{2}\right)^j$$

Hom 6

a) $X_n = \begin{cases} 1, \text{bark} \\ -1, \text{death} \end{cases}$

$P(X_n = 1) = P(\text{bark}) = 3/4$

$P(X_n = -1) = P(\text{death}) = 1/4$

$S = \{0, \pm 1, \pm 2, \pm 3, \dots\}$

$S_n, n = \{0, 1, 2, 3, \dots\}$

$S_n = X_1 + X_2 + X_3 + \dots + X_n$

b) $P_{ij} = (S_{n+1} = j | S_n = i)$
 $= P(X_{n+1} = j - i)$

$\Rightarrow \begin{cases} 3/4, j-i = 1 \\ 1/4, j-i = -1 \\ 0, \text{otherwise} \end{cases}$

c) $P(X_1 = -1, X_2 = 1, X_3 = -1)$

$\Rightarrow P(X_1 = -1) P(X_2 = 1) P(X_3 = -1)$

$\Rightarrow 1/4 \cdot 3/4 \cdot 1/4$

$\Rightarrow 3/64$

d) $P(X_1 = -1, X_2 = -1, X_3 = -1, X_4 = -1)$

$\Rightarrow P(X_1 = -1) P(X_2 = -1) P(X_3 = -1) P(X_4 = -1)$

$\Rightarrow 1/4 \cdot 1/4 \cdot 1/4 \cdot 1/4$

$\Rightarrow 1/256$

e) Net amount of 2 death by day 4

$\Rightarrow 3 \text{ bark} \& 1 \text{ death}$

$P(\text{bark}) = 3/4, P(\text{death}) = 1/4$

$(3/4 \cdot 1/4 \cdot 1/4 \cdot 1/4) \times 4$

$\Rightarrow 3/64$

Hom 7

a) $X_n = \begin{cases} 1, 2, 3, \text{bark} \\ 0, \text{no bark/death} \\ -1, -2, -3, \text{death} \end{cases} \quad n = 0, 1, 2, 3, \dots$

$S = \{0, \pm 1, \pm 2, \pm 3, \dots\}$

b) $P(X_n = 1) = 0.3$

$P(X_n = 2) = 0.1$

$P(X_n = 3) = 0.1$

$P(X_n = -1) = 0.2$

$P(X_n = -2) = 0.1$

$P(X_n = -3) = 0.1$

$P_{ij} = \begin{cases} 0.1, j = i+3 \\ 0.1, j = i+2 \\ 0.2, j = i+1 \\ 0.3, j = i \\ 0.3, j = i-1 \\ 0.1, j = i-2 \\ 0.1, j = i-3 \\ 0, \text{otherwise} \end{cases}$

c) $P(X_1 = -3, X_2 = 1, X_3 = -1, X_4 = 0)$

$\Rightarrow P(X_1 = -3) P(X_2 = 1) P(X_3 = -1) P(X_4 = 0)$

$\Rightarrow (0.1)(0.3)(0.2)(0.1)$

$\Rightarrow 0.0006$

d) $P(X_1 = -1, X_2 = -1, X_3 = -1, X_4 = -1)$

$\Rightarrow P(X_1 = -1) P(X_2 = -1) P(X_3 = -1) P(X_4 = -1)$

$\Rightarrow (0.2)^4 \Rightarrow 0.0016$

e) $P(S_4 = -2)$

Net amount of 2 death means net change is -2 after 4 days

$P(\sum_{i=1}^4 X_i = -2)$

$X_i \in \{-3, -2, -1, 0, 1, 2, 3\}$

less = $E(S^2)$

How 8

① X_n : No of deer on field B on day n day

$$X_n = 0, 1, 2, \dots, 580$$

$$\{X_n, n \geq 0, 1, 2, 3, \dots\}$$

X_0 : Number of initial deer on field B = 100

State Space $S = \{0, 1, 2, 3, \dots, 580\}$

$$P_{ij} = P(X_{n+1} = j | X_n = i)$$

$$= P(X_{n+1} = j - i)$$

$$= \begin{cases} i - \frac{1}{2}, & j = i+1 \\ \frac{i}{2}, & j = i-1 \end{cases}$$

or otherwise

$$P_{i,i+1} = 1 - i/580$$

$$P_{i,i-1} = i/580$$

$$② X_0 = 100, X_1 = 99, X_2 = 98, X_3 = 97$$

$$P(X_3 = 97 | X_2 = 98) P(X_2 = 98 | X_1 = 99)$$

$$\Rightarrow P(X_2 = 98 | X_1 = 100)$$

$$\Rightarrow P_{100,99} P_{99,98} P_{98,97}$$

$$\Rightarrow \left(1 - \frac{99}{580}\right) \left(\frac{99}{580}\right) \left(\frac{100}{580}\right)$$

$$\Rightarrow \left(\frac{481}{580}\right) \left(\frac{99}{580}\right) \left(\frac{100}{580}\right)$$

How 9

$$① P(X_4 = 301, X_3 = 300, X_2 = 301)$$

$$\Rightarrow \sum_{j=0}^{580} P(X_4 = 301 | X_3 = 301) P(X_3 = 300 | X_2 = 301) P(X_2 = 301 | X_1 = k) P(X_1 = k | X_0 = j) P(X_0 = j)$$

$$\Rightarrow \sum_{j=0}^{580} P_{300,301} P_{301,300} P_{k,301} P_{j,k} \pi_0(j)$$

$$\Rightarrow \left(1 - \frac{300}{580}\right) \left(\frac{301}{580}\right) \frac{1}{581} \sum_{j=0}^{580} P_{k,301} P_{j,k} \pi_0(j)$$

$$\Rightarrow \frac{280}{580} \cdot \frac{301}{580} \cdot \frac{1}{581} \sum_{j=0}^{580} P_{j,301}^2$$

$$② P(X_2 = 300)$$

$$= \sum_{j=0}^{580} P(X_2 = 300 | X_1 = k) P(X_1 = k | X_0 = j) P(X_0 = j)$$

$$\Rightarrow \sum_{j=0}^{580} P_{k,300} P_{j,k} \pi_0(j) \quad \boxed{P_{300,300}^2 = P_{300,299} P_{299,300} + P_{300,301} P_{301,300}}$$

$$\Rightarrow \frac{1}{581} \sum_{j=0}^{580} P_{j,300}^2$$

$$\Rightarrow \frac{1}{581} \sum_{j=0}^{580} P_{j,300}^2 = \frac{1}{581} [P_{300,300}^2 + P_{300,299}^2 + P_{302,300}^2]$$

$$P_{299,300}^2 = P_{299,299} P_{299,300} \quad P_{302,300}^2 = P_{302,301} P_{301,300}$$

HW10

- ② X_n = Gambler's Capital at time n
 X_n = (Number won up to n) - (Number of losses up to n)
 $X_0 = 0$
 $P(\text{Win}) = P(Y_n = 1) = 9/19$; $P(\text{Loss}) = P(Y_n = -1) = 10/19$
 Quits when $X_n = -10$ or $X_n = +25$

B) $S = \{-10, -9, \dots, 24, 25\}$

C) $P_{i,i+1} = 9/19$
 $P_{i,i-1} = 10/19$
 $P_{ij} = 0$, otherwise
 $P_{-10,-10} = P_{25,25} = 1$

D) $-10 \neq 25$

E) $P(Y_1 = -1, Y_2 = 1, Y_3 = 1, Y_4 = -1)$
 $X_0 = 0, X_1 = 0 + Y_1 = -1, X_2 = X_1 + Y_2 = 0$
 $X_3 = X_2 + Y_3 = 1, X_4 = X_3 + Y_4 = 0$
 $P(X_4 = 0 | X_3 = 1) P(X_3 = 1 | X_2 = 0) P(X_2 = 0 | X_1 = -1) \cdot P(X_1 = -1 | X_0 = 0)$

$\Rightarrow P_{0,-1} P_{-1,0} P_{0,1} P_{1,0} = \frac{10}{19} \cdot \frac{9}{19} \cdot \frac{9}{19} \cdot \frac{10}{19} = \frac{8100}{130321}$

F) $P(X_2 = -1) = 10/19$ Since the bets are iid.

HW11

Given:

$P(\text{Win}) = 2/3$
 $P(\text{Loss}) = 1/3$
 $P_{i,i+1} = 2/3$
 $P_{i,i-1} = 1/3$
 $P_{ij} = 0$, otherwise

② $P(X_1 = 1, X_2 = -1, X_3 = -1, X_4 = 1)$
 $\Rightarrow P(X_1 = 1) P(X_2 = -1) P(X_3 = -1) P(X_4 = 1)$
 $\Rightarrow \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3}$
 $\Rightarrow 4/81$

③ $P(T_{\{2,8\}}) = 6$

Quits in sixth bet $P_{22} = P_{44} = 1$

Wets 8 with 6 bets $\Rightarrow$ Net Jan +4 $\Rightarrow$ 5W & 1L
 Wets 2 with 6 bets $\Rightarrow$ Net Loss -2 $\Rightarrow$ 4L & 2W

4	L	W	W	W	W	W
	W	L	W	W	W	W
	W	W	L	W	W	W
	W	W	W	L	W	W
	W	W	W	W	L	W
	W	W	W	W	W	L

$P(W) = 2/3$

$P(L) = 1/3$

$(\frac{2}{3})^5 \cdot \frac{1}{3} \cdot 4 = \frac{128}{3^6} = \frac{128}{729}$

W L L L L L

$(\frac{1}{3})^4 (\frac{2}{3})^2 \times 5 = \frac{20}{729}$

$\Rightarrow \left(\frac{128 + 20}{729} \right) = \frac{148}{729}$

$$P(X_{n+1} = 3 | X_n = 1) = \frac{1}{4} = p_{13}$$

$$P(X_{n+1} = 1 | X_n = 1) = \frac{3}{4} = P_{1,1}$$

$$P(X_{n+1} = 1 | X_n = 3) = \frac{2}{3} = p_{31}$$

$$P(X_{n+1}=3 | X_n=3) = 1/3 = p_{33}$$

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} 3/4 & 1/4 \\ 2/3 & 1/3 \end{pmatrix}$$

$$\textcircled{b} \begin{pmatrix} 3/4 & 1/6 \\ 2/3 & 1/3 \end{pmatrix} \begin{pmatrix} 3/4 & 1/4 \\ 2/3 & 1/3 \end{pmatrix} = \begin{pmatrix} 3/16 + 1/6 & 3/16 + 1/12 \\ 1/2 + 2/9 & 1/6 + 1/9 \end{pmatrix}$$

$$P_{13} = 13/48$$

$$P(X_{n+2}=3/X_n=1) = P_{13}^2 = P \cdot P_{(1,3)} \text{ or } P_{(3,1)}$$

13/48

$$P(X_{n+2}=1 | X_{n+1}=1) = p_{11}^2 = p \cdot p_{(1,1)} \text{ etc.}$$

$$\Rightarrow \frac{9}{16} + \frac{1}{6} = \frac{35}{48}$$

(c) $P(X_{0.15} = 3 | X_0 = 1)$ or $P(X_{0.15} = 3 | X_0 = 3)$

$$P(X_6)P_{13}^5 + P(X_6)P_{32}^5$$

$$= 0.5(P_{13}^5 + P_{32}^5)$$

$$P(X_{0.75} = 21 | X_0 = 21) \text{ or } P(X_{0.75} = 21 | X_0 = 9)$$

$$P(X_0)P_{\mu}^5 + P(X_0)P_{\mu}^5$$

$$\rightarrow 0.5 \left(p_{14}^S + p_{21}^S \right)$$

(c) $P_{i+1} = 1/2$

$$P_{n-1} = 1/2$$

Page 21

$$P_{LL} = 1$$

$$S = \{0, 1, 2, 3, 4, 5, 6\}$$

$$p(T_5 = 2^{nd} \text{ bet}) = p(T_5 = 2)$$

$$K_2 = 5, K_1 \neq 5, K_0 = 3$$

$$P(X_1 \neq 5, X_2 = 5 | X_0 = 3)$$

$$X_0 = 3 \begin{cases} \alpha_1 = 4 \\ \alpha_1 = 2 \end{cases}$$

$$P(X_1=4, X_2=5 | X_0=3) = P(X_2=5 | X_1=4) P(X_1=4 | X_0=3) P(X_0=3)$$

$$\Rightarrow P_{45} P_{34} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

(b) $P(T_4=3 | X_0=3)$

⇒ The first time gambler's Capital is p in the 3rd bet

$$P(X_0=4 | X_0=3) = P(X_1 \neq 4, X_2 \neq 4, X_3=4 | X_0=3)$$

$$X_0 = 3 \begin{matrix} x_1 = 4 \\ x_1 = 2 \end{matrix} = P(X_1 = 2, X_2 = 4, X_3 = 4 \mid X_0 = 3)$$

$$X_1=2 \begin{cases} X_2=3 \\ X_2=1 \end{cases} = P(X_1=2, X_2=3, X_3=4 | X_0=3)$$

$$\Rightarrow P(X_3=4|X_2=3) P(X_2=3|X_1=2) \cdot P(X_1=2|X_0=3) P(X_0=3)$$

$$\rightarrow p_1 \cdot p_2 \cdot p_3$$

$$\Rightarrow \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

$$(c) P(T_4 \leq 3)$$

2) $T_4 = 1$ or $T_4 = 2$ or $T_4 = 3$ | X_{023}

$$P(X_1=4) = P(X_2=4) = P(X_3=4) = 1/8 \text{ (from part 3)}$$

$X_1 \begin{matrix} \swarrow 4 \\ \searrow 2 \end{matrix} \begin{matrix} \swarrow 5 \\ \searrow 3 \end{matrix} \Rightarrow P(X_2=4) = 0$

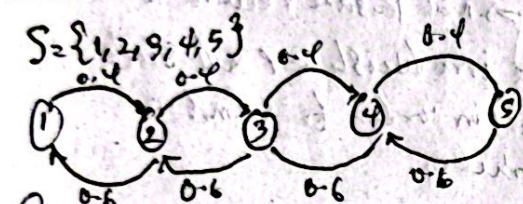
$$\Rightarrow P(T_4 \leq 3) = P(X_1=4) + P(X_2=4)$$

$$\Rightarrow P(X_1=4 | X_0=3) + 1/8$$

$$\Rightarrow P_{34} = P(X_0=3) + 1/8$$

$$\Rightarrow \frac{1}{2} + \frac{1}{8} = \frac{4+1}{8} = \frac{5}{8}$$

Homework 5.1



② $\{1, 2, 3, 4, 5\}$ - The only class

③ None

④ None

⑤ All states transient = 1, 2, 3, 4, 5

⑥ Yes. Has only one class $\{1, 2, 3, 4, 5\}$

Homework 5.2

$S = \{0, 1, 2, 3, 4, 5, 6\}$

① $\{0\}, \{1, 2, 3\}, \{4, 5, 6\}$

② None $P_{ij} = 1$ = Absorbing state

③ Closed classes

$\{1, 2, 3\}, \{4, 5, 6\}$

④ Recurrent class = closed class

Recurrent class state = 1, 2, 3, 4, 5, 6

Transient state = 0

⑤ Chain not irreducible Has 3 disjoint classes

Homework 6.1

① $\{X_n, n=0, 1, 2, 3, \dots\}$

$S = \{1, 3\}$

$$P(X_{n+1}=9 | X_n=1) = \frac{2}{3} = p_{13}$$

$$P(X_{n+1}=1 | X_n=1) = \frac{1}{3} = p_{11}$$

$$P(X_{n+1}=1 | X_n=3) = \frac{3}{5}$$

$$P(X_{n+1}=3 | X_n=3) = \frac{2}{5}$$

$$P_2 = \begin{pmatrix} p_{11} & p_{13} \\ p_{31} & p_{33} \end{pmatrix} = \begin{pmatrix} 1/3 & 2/3 \\ 3/5 & 2/5 \end{pmatrix}$$

② $\pi_i = \sum_j \bar{\pi}_j p_{ji}$ $i=1, 2, 3$

$$\bar{\pi}_1 = \sum_j \bar{\pi}_j p_{j1} = \bar{\pi}_1 p_{11} + \bar{\pi}_3 p_{31}$$

$$= \bar{\pi}_1 \cdot \frac{1}{3} + \bar{\pi}_3 \cdot \frac{3}{5}$$

$$\bar{\pi}_1 = \frac{\bar{\pi}_3}{3} + \frac{3}{5} \bar{\pi}_3$$

$$\frac{2}{3} \bar{\pi}_1 = \frac{3}{5} \bar{\pi}_3$$

$$\bar{\pi}_1 = \frac{9}{10} \bar{\pi}_3$$

③ $\sum_j \bar{\pi}_j = 1$

$$\bar{\pi}_1 + \bar{\pi}_3 = 1$$

By (i) & (ii)

$$\frac{9}{10} \bar{\pi}_3 + \bar{\pi}_3 = 1$$

$$\frac{19}{10} \bar{\pi}_3 = 1 \Rightarrow \bar{\pi}_3 = \frac{10}{19}$$

$$\bar{\pi}_1 = 1 - \frac{10}{19} = \frac{9}{19}$$

$$\bar{\pi} = (\bar{\pi}_1, \bar{\pi}_3)$$

$$= \left(\frac{9}{19}, \frac{10}{19} \right)$$

Hom 6.2

1) $\{X_n, n=0,1,2,3,\dots\}$

2) $\{X_n, n=0,1,2,3,\dots\}$

X_n = No of deer passing a fixed B on n^{th} day

$S = \{0, 1, 2, \dots, 580\}$

$d = 580$

$P_{i,i+1} = 1 - \frac{1}{d}$

$P_{i,i-1} = \frac{1}{d}$

$P_{ij} = 0$, otherwise

3) $\pi_j = \sum_{i=0}^{\infty} \pi_i P_{ij}$ for $j \in S$

$\pi_{290} = \sum_{i=0}^{580} \pi_i P_{i,290} \Rightarrow 290 \pi_{290} = 301 \pi_{291}$

$\Rightarrow \pi_{291} P_{291,290} + \pi_{300} P_{300,290}$

$\Rightarrow \frac{291}{580} \pi_{291} + \frac{301}{580} \pi_{301}$

$\pi_{301} = \pi_{300} P_{300,301} + \pi_{291} P_{291,301}$

$\Rightarrow \frac{290}{580} \pi_{300} + \frac{302}{580} \pi_{302}$

$\pi_{302} = \frac{291}{580} \pi_{301} + \frac{303}{580} \pi_{303}$

$\Rightarrow \pi_j = \left(\frac{1}{d}\right) \pi_0 \cdot j$ for $j=0,1,2,\dots,580$

$\sum_j \pi_j = 1$

$\sum_j \left(\frac{1}{d}\right) \pi_0 j = 1$

$\left(\frac{1}{d}\right) \pi_0 \sum_{j=0}^d j = 1$

$(2) \pi_0 = 1$

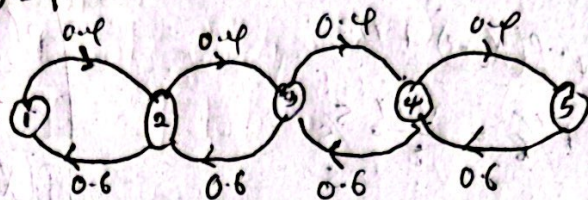
$\pi_0 = \frac{1}{2}$

$\pi_j = \left(\frac{1}{j}\right) \frac{1}{2}$ for $j=0,1,2,\dots$

$\pi_j = \left(\frac{1}{j}\right) \frac{1}{2^{580}}$ for $j=0,1,2,3,\dots$

Homework 6.3

$S = \{1, 2, 3, 4, 5\}$ finite



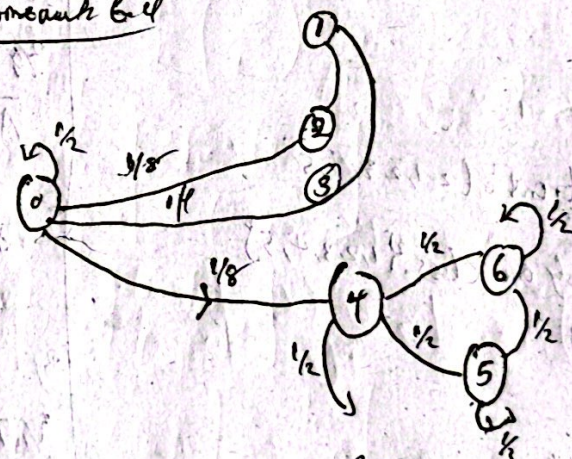
NO!

Although the Markov chain is irreducible it is not closed hence hence transient

- 1 & 5 are not closed $\Rightarrow$ not positive recurrent

- A finite MC must be irreducible at all states must be positive recurrent in order to admit a unique stationary distribution.

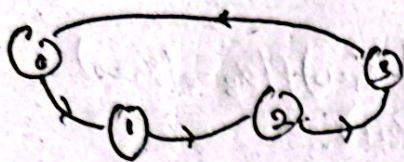
Homework 6.4



NO! The MC not irreducible and is closed class $\{1,2,3\}$ & $\{4,5,6\} \Rightarrow 2$ a MC stationary distribution.

No-Unique stationary distribution on full state space

Homework 7.1



Class = {0, 1, 2, 3}

⇒ All other states have period 4 since they belong to same class as 0.

Homework 7.2

$$P_{i,i+1} = 1 - \frac{1}{2}$$

$$S = \{0, 1, 2, \dots, d\}$$

$$P_{0,0} = \frac{1}{2}$$

Stochastic distribution

$$\textcircled{1} \bar{\pi}_0 = \sum_{i,j} \bar{\pi}_i P_{ij} \quad i,j \in S$$

$$\textcircled{2} \sum_i \bar{\pi}_i = 1$$

$$j=0 \quad \bar{\pi}_1 = \sum_i \bar{\pi}_i P_{i1}$$

$$\bar{\pi}_0 = \bar{\pi}_1 P_{10}$$

$$= \bar{\pi}_1 \frac{1}{2}$$

$$\Rightarrow \bar{\pi}_1 = 2\bar{\pi}_0$$

$$j=1 \quad \bar{\pi}_1 = \sum_i \bar{\pi}_i P_{i1}$$

$$\Rightarrow \bar{\pi}_0 P_{01} \neq \bar{\pi}_1 P_{11}$$

$$\bar{\pi}_1 = \bar{\pi}_0 + \frac{1}{2} \bar{\pi}_2$$

$$\bar{\pi}_2 = \frac{1}{2} (d-1) \bar{\pi}_0$$

$$\bar{\pi}_d = \left(\frac{1}{2}\right)^d \bar{\pi}_0$$

$$\sum \left(\frac{1}{2}\right)^i \bar{\pi}_0 = 1$$

$$\bar{\pi}_0 = \frac{1}{2^d}$$

$$\bar{\pi}_i = \binom{d}{i} \left(\frac{1}{2}\right)^d \quad i=0,1,2,\dots \quad \{\text{stochastic distribution}\}$$

⇒ The MC has period 2 since from state i you always move to $i+1$ so you alternate parity every step

$P_{0,0} > 0$ - never occur

$P_{0,0} > 0$ - always

$$P_{ii} > 0 \Leftrightarrow n \text{ is even}$$

Since MC is not aperiodic $d \neq 1$, the limiting distribution does not exist.

Homework 7.4

$$S = \{1, 2, 3\}$$

Ⓐ Yes! Ⓐ final state space

Ⓑ Existence of self-loops ⇒ aperiodic

$$\textcircled{b} \bar{\pi}_i = \sum_j \bar{\pi}_j P_{ji}$$

$$\bar{\pi} = \sum_i \bar{\pi}_i P_{i1}$$

$$= \bar{\pi}_2 P_{21} + \bar{\pi}_1 P_{11}$$

$$\Rightarrow \bar{\pi}_2 (0.2) + \bar{\pi}_1 (0.6)$$

$$\bar{\pi}_1 = \bar{\pi}_2 (0.2) + \bar{\pi}_1 (0.6)$$

$$(0.4) \bar{\pi}_1 = \bar{\pi}_2 (0.2)$$

$$\bar{\pi}_2 = 2\bar{\pi}_1$$

$$\bar{\pi}_3 = \sum_i \bar{\pi}_i P_{i3}$$

$$\Rightarrow \bar{\pi}_2 P_{23} + \bar{\pi}_3 P_{33} \Rightarrow \bar{\pi}_3 (0.3) + \bar{\pi}_3 (0.2)$$

$$0.5 \bar{\pi}_3 = 0.3 \bar{\pi}_2$$

$$\bar{\pi}_3 = \frac{3}{5} \bar{\pi}_2$$

$$\bar{\pi}_2 = 2\bar{\pi}_1 \quad \bar{\pi}_3 = \frac{6}{5} \bar{\pi}_1$$

$$\sum \bar{\pi}_i = 1 \Rightarrow \bar{\pi}_1 + \bar{\pi}_2 + \bar{\pi}_3 = 1$$

$$\bar{\pi}_1 + 2\bar{\pi}_1 + \frac{6}{5} \bar{\pi}_1 = 1$$

$$\bar{\pi}_1 = \left(\frac{1}{4}, \frac{2}{4}, \frac{6}{4}\right)$$

Homework 8.1

$$Q) P_2 \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix} = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}$$

$$P(T_1 = m | X_0 = 1) = ?$$

$$S = \{0, 1\}$$

$$m = 0, 1, 2, 3, \dots$$

$$m=0, P(T_1=0 | X_0=1) = 1$$

$$m=1$$

$$P(T_1=1 | X_0=1) = P(X_1=1 | X_0=1) = p_{11}$$

$$m=2$$

$$P(T_1=2 | X_0=1) = P(X_2=1 | X_0=0) P(X_1=0 | X_0=1) \\ \Rightarrow p_{01} p_{10}$$

$$P(T_1=m | X_0=1) = p_{01} (p_{00})^{m-2} \quad m=2, 3, 4, \dots$$

$$P(T_1=1 | X_0=1) = p_{11}$$

$$N_1 = E[T_1 | X_0=1]$$

$$\Rightarrow p_{11} + p_{01} p_{10} + p_{01} p_{00} p_{10} + p_{01} (p_{00})^2 + \dots$$

$$\Rightarrow p_{11} + \sum_{n=2}^{\infty} n p_{01} (p_{00})^{n-2} p_{10}$$

$$\Rightarrow \sum_{n=2}^{\infty} n p_{01} (p_{01})^{n-2} p_{10} = p_{01} p_{10} \sum_{n=2}^{\infty} n (p_{00})^{n-2}$$

$$= p_{01} p_{10} \cdot \sum_{n=2}^{\infty} n (p_{00})^{n-2} \cdot \frac{p_{00}}{p_{00}} = \frac{p_{01} p_{10}}{p_{00}} \sum_{n=2}^{\infty} n (p_{00})^{n-1}$$

$$\text{now } \frac{d}{dx} x^n = n \cdot x^{n-1}, \quad x = p_{00}$$

$$N_3$$

$$P(T_1=3 | X_0=1) = P(X_3=1 | X_2=0) P(X_2=0 | X_1=0) P(X_1=1 | X_0=1) \\ \Rightarrow p_{01} p_{00} p_{10}$$

$$\frac{p_{01} p_{10}}{p_{00}} \sum_{n=2}^{\infty} \frac{d}{dx} x^n = \frac{p_{01} p_{10}}{p_{00}} \frac{d}{dx} \sum_{n=2}^{\infty} x^n$$

$$M.P.$$

$$P(T_1=4 | X_0=1) = P(X_4=1 | X_3=0) P(X_3=0 | X_2=0) P(X_2=1 | X_1=0) P(X_1=1 | X_0=1) \\ \Rightarrow p_{01} p_{00} p_{00} p_{10}$$

$$m=4$$

$$P(T_1=5 | X_0=1) = p_{01} p_{00} p_{00} p_{00} p_{10}$$

$$P(T_1=m) = p_{01} (p_{00})^{m-2} p_{10} \quad m=2, 3, 4$$

$$P(T_1=1 | X_0=1) = p_{11}$$

$$P_2$$

$$\Rightarrow \frac{p_{01} p_{10}}{p_{00}} \frac{d}{dx} \left(\frac{x^2}{1-x} \right) = \frac{p_{01} p_{10}}{p_{00}} \left[\frac{2x - x^2}{(1-x)^2} \right]$$

$$\Rightarrow \frac{p_{01} p_{10}}{p_{00}} \left[\frac{2p_{00} - (p_{00})^2}{(1-p_{00})^2} \right]$$

$$\Rightarrow p_{01} p_{10} \left[\frac{2-p_{00}}{(1-p_{00})^2} \right]$$

$$N_1 = p_{11} + p_{01} p_{10} \left[\frac{2-p_{00}}{(1-p_{00})^2} \right]$$

$$\sum_{n=2}^{\infty} x^n = G.S$$

$$\sum_{n=2}^{\infty} x^n = \frac{x^2}{1-x} \quad x < 1$$

$$\text{Recall: } P_2 \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix} = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}$$

$$N = 1-p + p \left[\frac{1+p}{1-p} \right]$$

$$\Rightarrow 1-p + \frac{p+p^2}{1-p} \Rightarrow \frac{1+p}{1-p}$$

Part of Homework chapter 4

Daniel Olofin

2025-06-19

Homework 9.1.1

Theoretical Framework: Gambler's Ruin Simulation

The Gambler's Ruin problem models a stochastic process in which a gambler places a sequence of bets, either gaining or losing one unit of capital at each step. This process is a **discrete-time Markov chain** defined on the finite state space:

$$S = \{0, 1, 2, \dots, d\}$$

where: - 0 and d are **absorbing states** (once reached, the process stops), - all other states are **transient**.

At each time step n , the gambler's capital X_n evolves according to the following transition probabilities:

$$P(X_{n+1} = i + 1 \mid X_n = i) = p, \quad P(X_{n+1} = i - 1 \mid X_n = i) = 1 - p$$

Here, p is the probability of winning a single bet, and $1 - p$ is the probability of losing. The process continues until the gambler's capital reaches either absorbing boundary.

Key Properties

- If $p = 0.5$, the process is a **symmetric random walk**.
- If $p > 0.5$, the gambler has a favorable edge and is more likely to reach the goal.
- If $p < 0.5$, the gambler is at a disadvantage and more likely to go bankrupt.

The **probability of eventual ruin**, when starting from capital i , is given by:

$$\mathbb{P}_{\text{ruin}}(i) = \begin{cases} \frac{1 - \left(\frac{p}{1-p}\right)^i}{1 - \left(\frac{p}{1-p}\right)^d}, & \text{if } p \neq 0.5 \\ 1 - \frac{i}{d}, & \text{if } p = 0.5 \end{cases}$$

This formula illustrates that even when $p > 0.5$, there is still a non-zero chance of going broke before reaching the goal.

Simulation Justification

In this project, we simulate **three sample paths** for a gambler with an initial capital of **\$3**, aiming to reach **\$100**. We explore two cases: - $p = 0.7$: a favorable game - $p = 0.3$: an unfavorable game

Each path is simulated using a nested loop, drawing outcomes from a **Bernoulli distribution** $B(1, p)$ to determine wins and losses at each step. If the gambler reaches an absorbing state (0 or 100), the simulation path terminates early by holding that value constant.

Part (a)

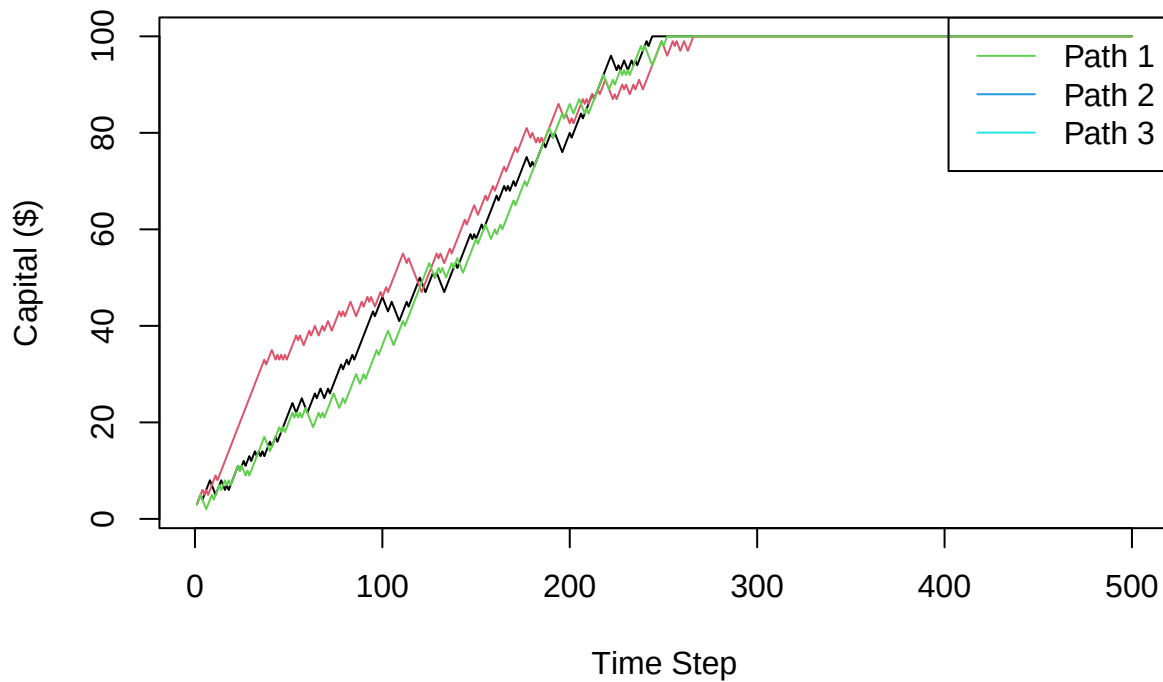
```
# PARAMETERS
d <- 100           # absorbing upper barrier
initialcap <- 3    # starting amount
time <- 500        # large enough to allow ruin or win
nsim <- 3          # 3 sample paths

# SIMULATE FOR p = 0.7 (Part a)
p <- 0.7
X <- matrix(0, nrow = time, ncol = nsim) # Capital
Y <- matrix(0, nrow = time, ncol = nsim) # Bernoulli trials
X[1, ] <- initialcap

for (i in 1:(time - 1)) {
  for (j in 1:nsim) {
    if (X[i, j] == 0 || X[i, j] == d) {
      X[i + 1, j] <- X[i, j] # Stay if absorbed
    } else {
      Y[i + 1, j] <- rbinom(1, 1, p)
      if (Y[i + 1, j] == 1) {
        X[i + 1, j] <- X[i, j] + 1
      } else {
        X[i + 1, j] <- X[i, j] - 1
      }
    }
  }
}

# PLOT for p = 0.7
matplot(1:time, X[, 1:3], type = "l", lty = 1, col = 1:3,
        ylab = "Capital ($)", xlab = "Time Step",
        main = "Gambler's Ruin: Sample Paths (p = 0.7)")
legend("topright", legend = paste("Path", 1:3), col = 3:6, lty = 1)
```


Gambler's Ruin: Sample Paths ($p = 0.7$)



part (b)

```
# PARAMETERS
d <- 100                # absorbing upper barrier
initialcap <- 3         # starting amount
time <- 500            # large enough to allow ruin or win
nsim <- 3               # 3 sample paths

# SIMULATE FOR p = 0.7 (Part a)
p <- 0.3
X <- matrix(0, nrow = time, ncol = nsim) # Capital
Y <- matrix(0, nrow = time, ncol = nsim) # Bernoulli trials
X[1, ] <- initialcap

for (i in 1:(time - 1)) {
  for (j in 1:nsim) {
    if (X[i, j] == 0 || X[i, j] == d) {
      X[i + 1, j] <- X[i, j] # Stay if absorbed
    } else {
      Y[i + 1, j] <- rbinom(1, 1, p)
      if (Y[i + 1, j] == 1) {
        X[i + 1, j] <- X[i, j] + 1
      } else {
        X[i + 1, j] <- X[i, j] - 1
      }
    }
  }
}
```

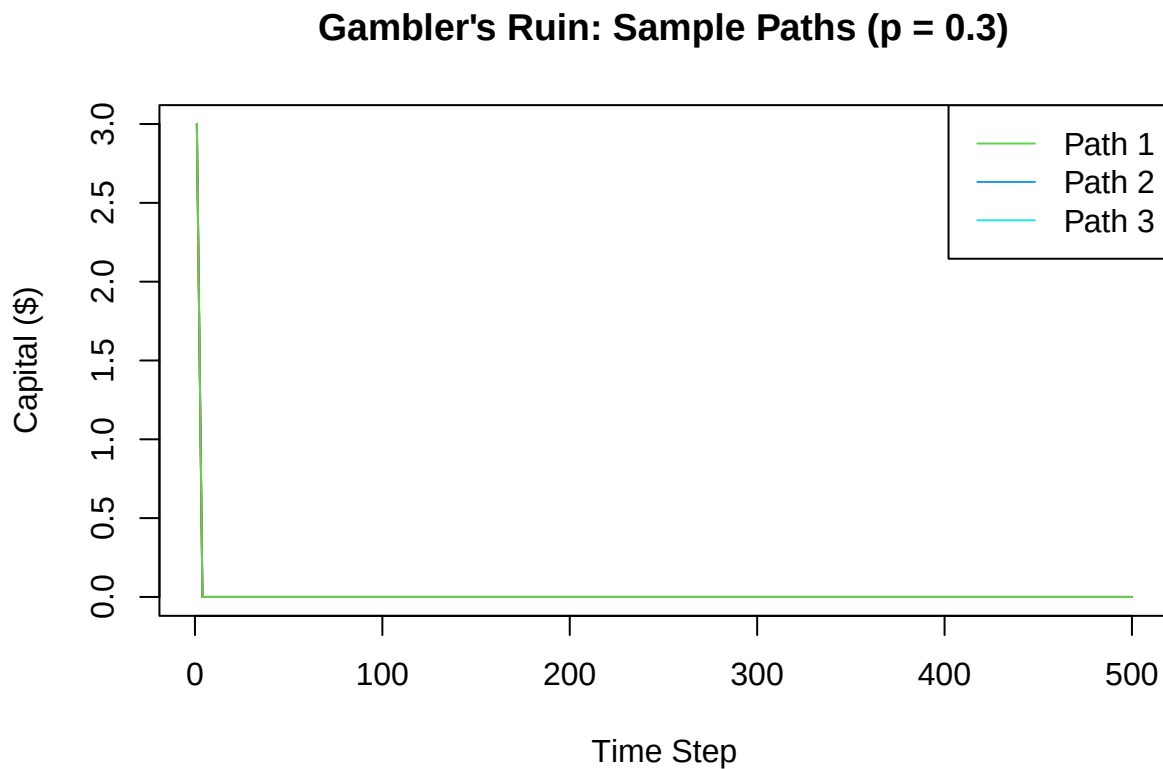


```

    }
  }
}

# PLOT for p = 0.3
matplot(1:time, X[, 1:3], type = "l", lty = 1, col = 1:3,
        ylab = "Capital ($)", xlab = "Time Step",
        main = "Gambler's Ruin: Sample Paths (p = 0.3)")
legend("topright", legend = paste("Path", 1:3), col = 3:6, lty = 1)

```



Homework 9.1.1 (Random Walk)

Theoretical Framework: Simple Random Walk Simulation

This simulation models a **simple random walk** — a discrete-time stochastic process in which a state moves up or down by 1 unit at each time step. The process is defined by the recurrence:

$$X_{n+1} = \begin{cases} X_n + 1 & \text{with probability } p \\ X_n - 1 & \text{with probability } 1 - p \end{cases}$$

The transition probabilities are:

$$P(X_{n+1} = i + 1 \mid X_n = i) = p, \quad P(X_{n+1} = i - 1 \mid X_n = i) = 1 - p$$

This defines a **time-homogeneous Markov Chain** on the set of integers, where transitions depend only on the current state and not the history of the process.

Key Properties

- If $p = 0.5$, the process is **symmetric** — equally likely to move up or down.
- If $p \neq 0.5$, the process has **drift** — it tends to move upward if $p > 0.5$, or downward if $p < 0.5$.
- There are **no absorbing states** unless explicitly added.

Simulation Plan

We simulate three independent sample paths of this random walk, starting from an initial value (e.g., 3), for a fixed number of steps (e.g., 100). We explore two different cases: - Case (a): $p = 0.5$ — symmetric walk
- Case (b): $p = 0.4$ — downward-biased walk

Each step of the walk is generated using a **Bernoulli trial** $B(1, p)$, where 1 indicates an upward move and 0 a downward move.

```
# Parameters
set.seed(123)
time <- 100      # number of steps
nsim <- 3        # number of sample paths
initialcap <- 3   # initial state

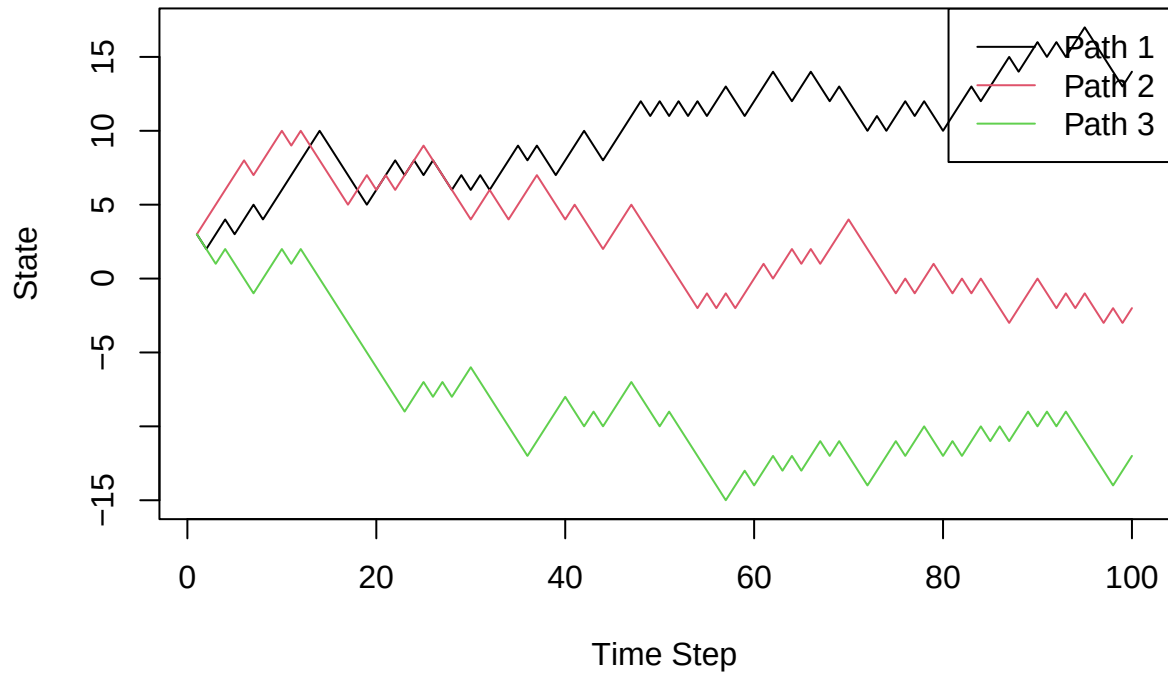
# CASE (a): p = 0.5 (Symmetric Walk)
p <- 0.5

X <- matrix(0, nrow = time, ncol = nsim) # Walk values
Y <- matrix(0, nrow = time, ncol = nsim) # Bernoulli outcomes
X[1, ] <- initialcap

for (i in 1:(time - 1)) {
  for (j in 1:nsim) {
    Y[i + 1, j] <- rbinom(1, 1, p)
    if (Y[i + 1, j] == 1) {
      X[i + 1, j] <- X[i, j] + 1
    } else {
      X[i + 1, j] <- X[i, j] - 1
    }
  }
}

# Plot for p = 0.5
matplot(1:time, X, type = "l", col = 1:nsim, lty = 1,
        ylab = "State", xlab = "Time Step", main = "Simple Random Walk (p = 0.5)")
legend("topright", legend = paste("Path", 1:nsim), col = 1:nsim, lty = 1)
```


Simple Random Walk ($p = 0.5$)



part (b)

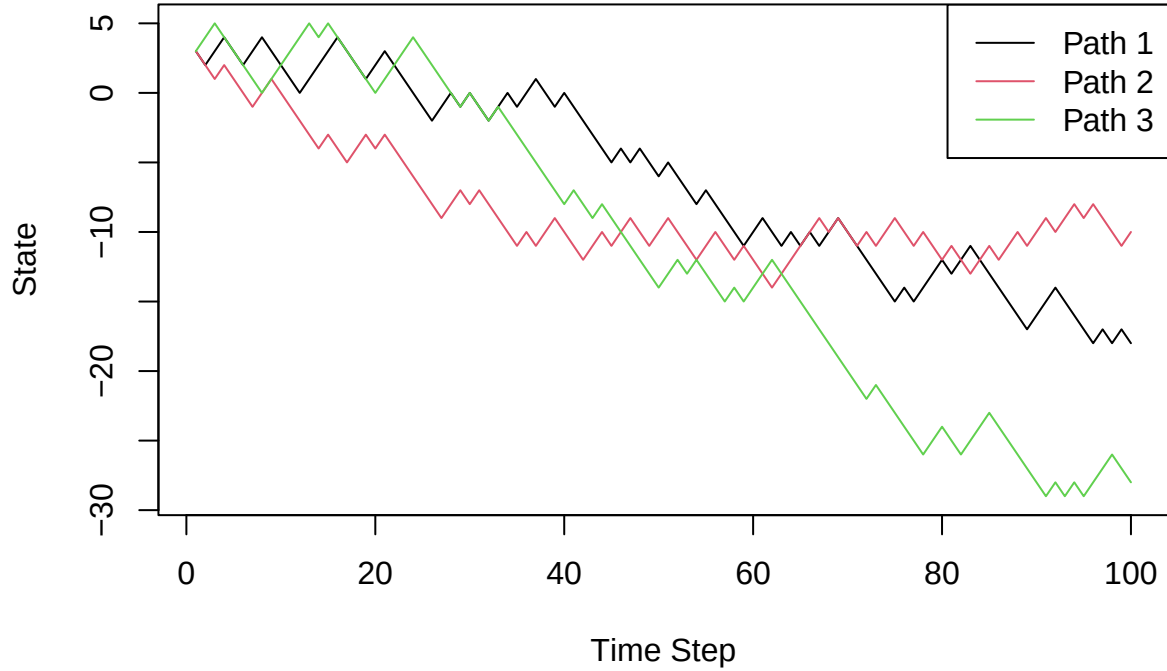
```
p <- 0.4

X <- matrix(0, nrow = time, ncol = nsim) # Walk values
Y <- matrix(0, nrow = time, ncol = nsim) # Bernoulli outcomes
X[1, ] <- initialcap

for (i in 1:(time - 1)) {
  for (j in 1:nsim) {
    Y[i + 1, j] <- rbinom(1, 1, p)
    if (Y[i + 1, j] == 1) {
      X[i + 1, j] <- X[i, j] + 1
    } else {
      X[i + 1, j] <- X[i, j] - 1
    }
  }
}

# Plot for p = 0.4
matplot(1:time, X, type = "l", col = 1:nsim, lty = 1,
        ylab = "State", xlab = "Time Step", main = "Simple Random Walk (p = 0.4)")
legend("topright", legend = paste("Path", 1:nsim), col = 1:nsim, lty = 1)
```


Simple Random Walk (p = 0.4)



Homework 9.1.2

Theoretical Framework: Random Walk Simulation and Inference

This exercise explores the behavior of a **simple random walk** through simulation and statistical inference. A simple random walk is a discrete-time stochastic process $\{X_n\}_{n \geq 0}$ defined by the recursive relationship:

$$X_{n+1} = \begin{cases} X_n + 1 & \text{with probability } p \\ X_n - 1 & \text{with probability } 1 - p \end{cases}$$

The process is **Markovian**, meaning the future state depends only on the present state, not the past history. The transition probabilities are:

$$P(X_{n+1} = i + 1 \mid X_n = i) = p, \quad P(X_{n+1} = i - 1 \mid X_n = i) = 1 - p$$

In this simulation, we fix $p = 0.4$, introducing a **negative drift**, which biases the walk downward over time. Each path begins at an initial state $X_0 = 3$ and is simulated for 20 steps.

Simulation and Sampling

A total of **200 independent random walk paths** are generated, each of length 20. For each path, the value of the process at the 20th step, X_{20} , is recorded. This yields a sample of 200 observations from the distribution of the random walk at time $n = 20$.

Inference: Confidence Intervals

The simulation allows us to estimate the **expected value** of the process at step 20. We compute a **sample mean** $\bar{X}_{20}$ and construct a confidence interval using the **Central Limit Theorem (CLT)**, which states that the sampling distribution of the sample mean is approximately normal for large sample sizes.

The **99% confidence interval** (for part a) and **95% confidence interval** (for part b) are computed using:

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

Where: - $\bar{x}$ is the sample mean of the 200 observations, - s is the sample standard deviation, - $n = 200$ is the sample size, - $z_{\alpha/2}$ is the critical value from the standard normal distribution.

This approach provides a statistically sound estimate for the **expected state of the process at step 20**, along with a measure of uncertainty.

```
# PARAMETERS
set.seed(123)
p <- 0.4           # probability of moving up
time <- 20         # simulate 20 steps
nsim <- 200        # number of simulations
initialcap <- 3    # starting state

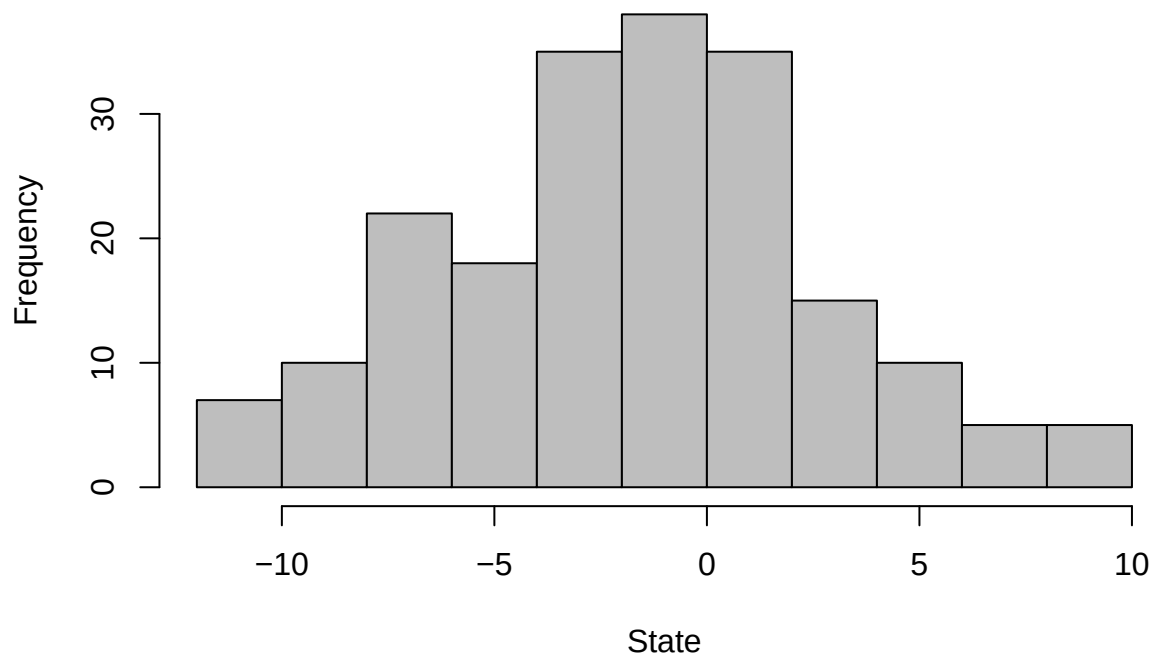
# STORAGE
X <- matrix(0, nrow = time, ncol = nsim) # random walk
Y <- matrix(0, nrow = time, ncol = nsim) # Bernoulli outcomes
X[1, ] <- initialcap

# SIMULATION LOOP (professor's approach)
for (i in 1:(time - 1)) {
  for (j in 1:nsim) {
    Y[i + 1, j] <- rbinom(1, 1, p)
    if (Y[i + 1, j] == 1) {
      X[i + 1, j] <- X[i, j] + 1
    } else {
      X[i + 1, j] <- X[i, j] - 1
    }
  }
}

# Extract state at 20th step
state_20 <- X[time, ]

# PART (a) - Histogram + Description
hist(state_20,
      col = "gray", main = "Histogram of State at Step 20",
      xlab = "State", ylab = "Frequency", breaks = 15)
```


Histogram of State at Step 20



```
cat("Shape: Slightly left-skewed\n")
```

```
## Shape: Slightly left-skewed
```

```
cat("Location: Mode around 0-2\n")
```

```
## Location: Mode around 0-2
```

```
cat("Dispersion: Spread from approx. -10 to 12\n")
```

```
## Dispersion: Spread from approx. -10 to 12
```

```
# PART (b) - 95% Confidence Interval for E[X20]
mean_20 <- mean(state_20)
sd_20 <- sd(state_20)
se_20 <- sd_20 / sqrt(nsim)
z <- qnorm(0.975) # for 95% CI

ci_95 <- c(mean_20 - z * se_20, mean_20 + z * se_20)
cat("Mean at step 20:", round(mean_20, 2), "\n")
```

```
## Mean at step 20: -0.73
```



```
cat("95% CI for E[X_20]: [", round(ci_95[1], 2), ",", round(ci_95[2], 2), "]\n")
```

```
## 95% CI for E[X_20]: [ -1.36 , -0.1 ]
```